48 hours of generation

Hourly forecast with a 95% interval, 27–28 October 2026

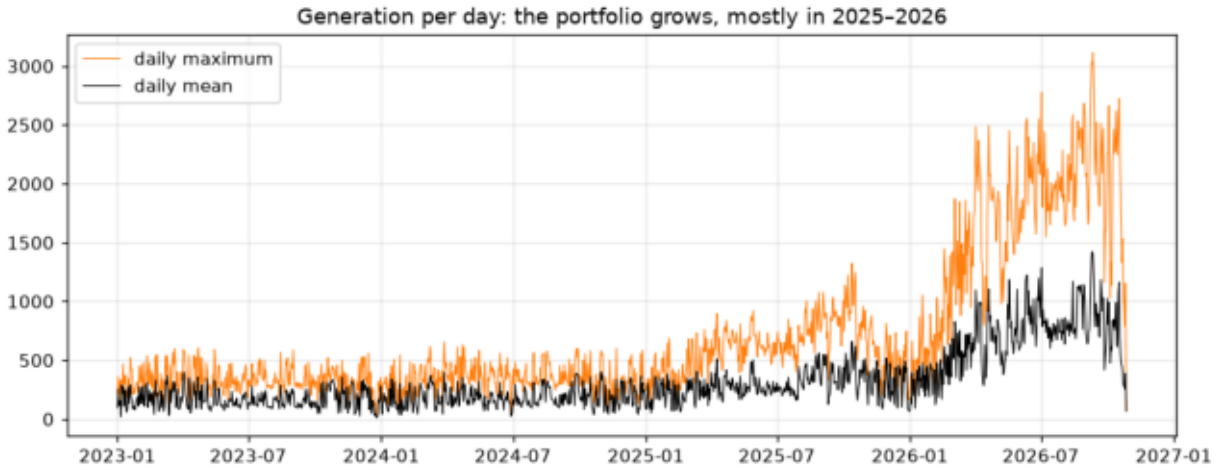
Artem Korniienko · MATH840 Time Series · KSE

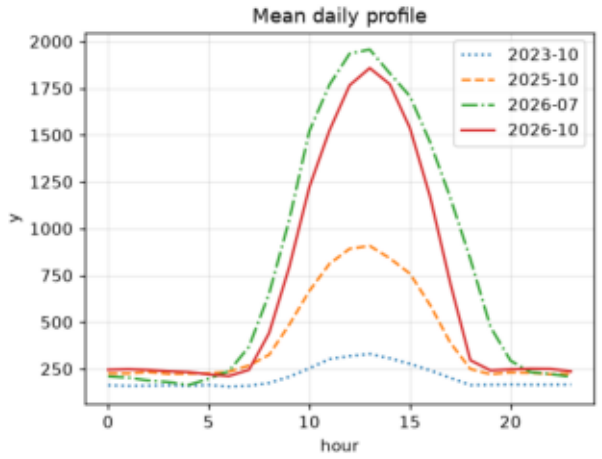
On 40 validation windows: RMSE −10%, MAPE −20%, Winkler −51% against seasonal naive

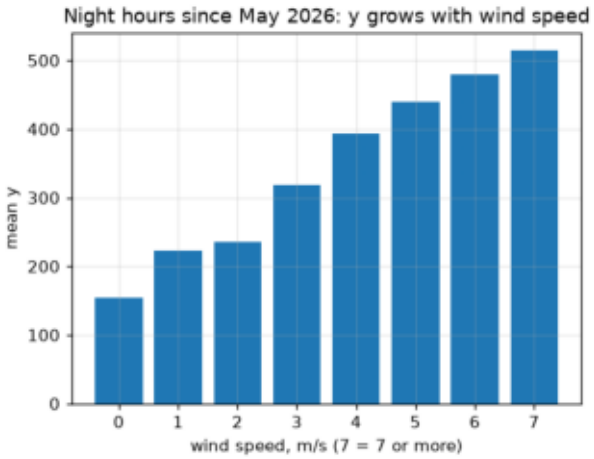
What is in the series

Solar by day, wind at night, and a portfolio that keeps growing

- The level grows: mean y 171 in 2023, 291 in 2025, 684 in 2026. The daily peak grew about 5 times.
- Daytime: a solar bell curve. At night: generation rises with wind speed.
- cloud_cover = 1 goes with the highest midday output, the opposite of its label (the series is disguised). I let the model learn the sign.
- The daylight flag moves one hour earlier at the end of October (clock change); the forecast already has 07–15.
- The weather forecast is rougher than real weather, more so on day 2.







Approach

Simple, and every feature is known 48 hours ahead

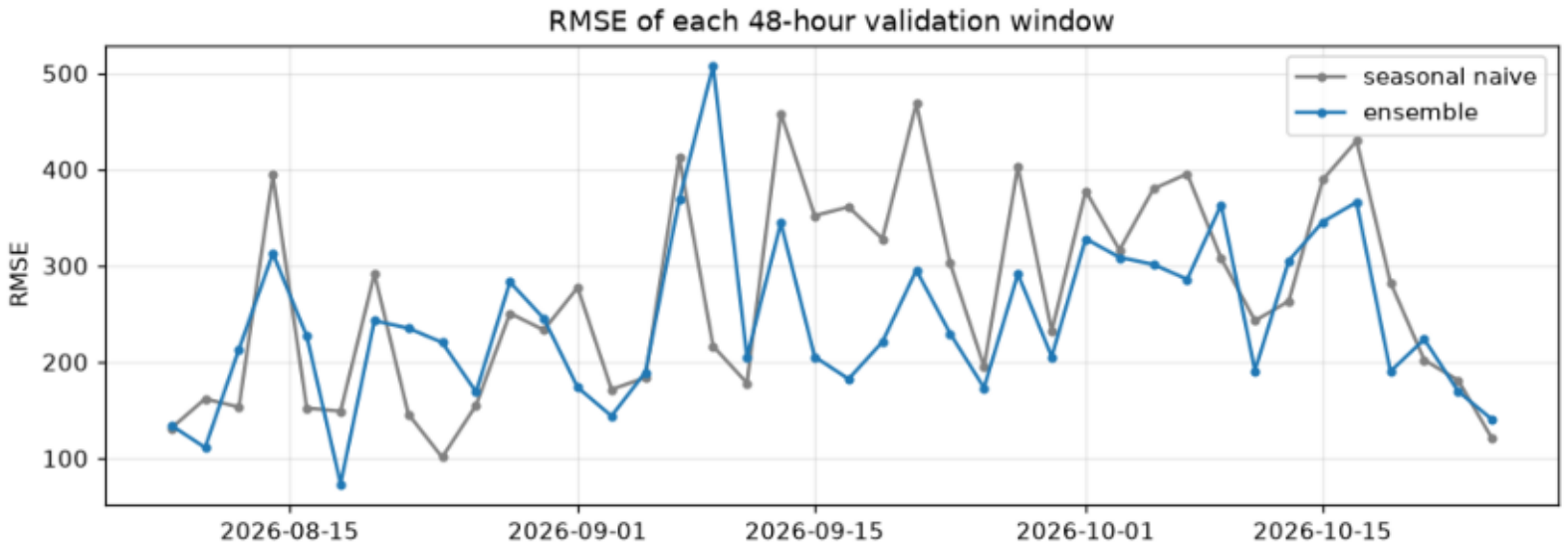
- Features: the 7 weather columns, hour, day of year, and last week's y at the same hour (mean, max, lag 48). The last week's level lets the model follow the growth.
- Two HistGradientBoosting models: one on y (good RMSE), one on log y (good MAPE). The forecast is their average.
- Validation: 40 rolling windows of 48 hours, each starting at midnight, every 2 days from 08 Aug to 25 Oct 2026. The model is refitted for every window.
- In validation the weather inputs get simulated forecast errors, larger on day 2, so the scores are not optimistic.
- 95% interval: the forecast plus the 2.5% and 97.5% quantiles of the validation errors, separately for day 1 / day 2 and daylight / night.

Validation against the benchmark

Mean over 40 windows. The average has the lower RMSE in 62% of them.

	RMSE	MAPE
Seasonal naive (benchmark)	269	153%
Mean of last 7 days	307	226%
GBM without weather	285	218%
GBM on y	235	149%
GBM on log y	264	97%
Average of the two GBMs	**243**	**123%**

- Day 1: level with the benchmark (RMSE 240 vs 237).
- Day 2, the private half: RMSE 235 vs 284, MAPE 77% vs 125%.
- Without weather the model loses to the benchmark.



How good is the interval

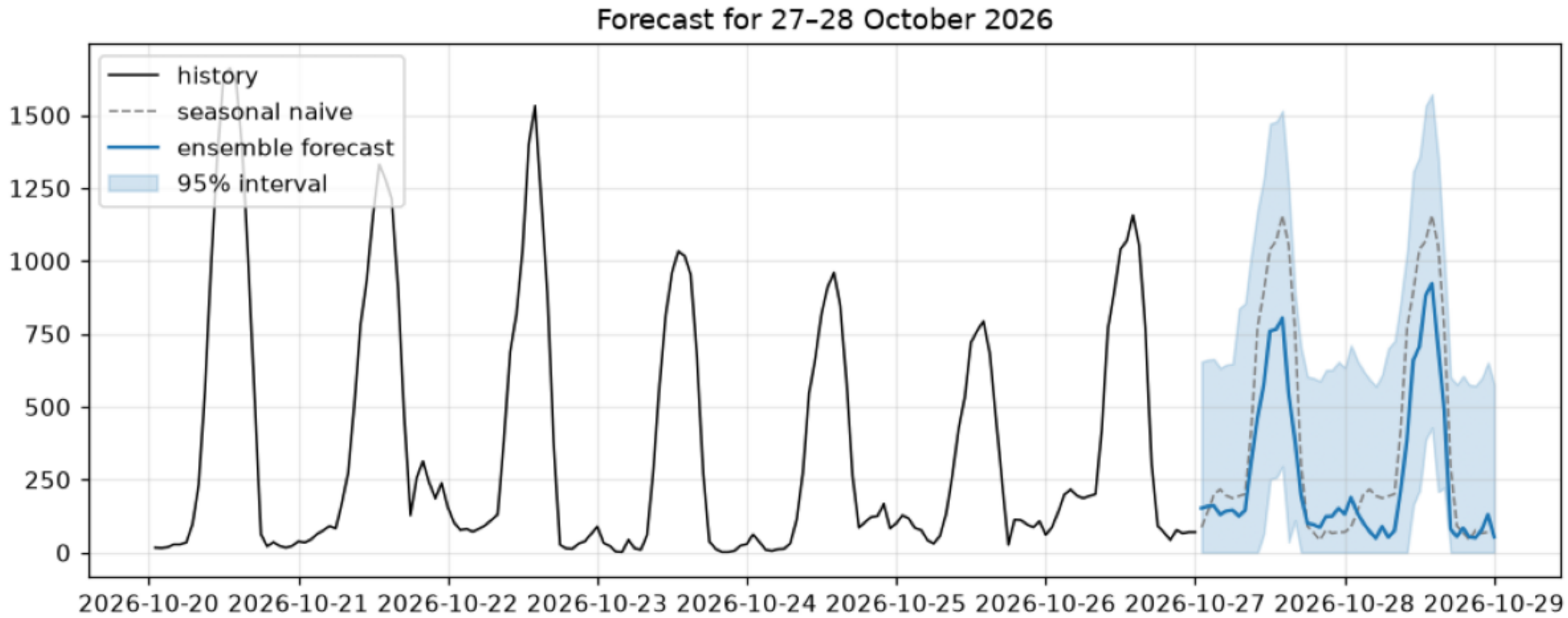
Calibrated on the first 20 windows, scored on the last 20

	Winkler	coverage
Average of the two GBMs	**1171**	**92%**
Seasonal naive (benchmark)	2397	75%

- Forecast + empirical 2.5% / 97.5% quantiles of past errors, by day 1 / day 2 and daylight / night; the lower bound is clipped at 0. No normality assumed: the errors are skewed upward.
- Tried first: quantiles of log(actual / forecast), so the width scales with the level. Worse Winkler and lower coverage: tiny night values make the ratios heavy-tailed.
- Coverage is a bit below 95% on the later windows: 96% in September but 90% in October. The final interval is calibrated on all 40 windows, the most recent included.

The forecast for 27–28 October

Day 1: humidity 94–100% and 5–10 °C, a low solar peak. Day 2: drier and warmer, a higher one.



What did not work, and what I would do next

- Mean of the last 7 days: worse than the benchmark (RMSE 307), because the weather moves solar output a lot from day to day.
- GBM on y alone: best RMSE (235) but MAPE 149%: it overshoots the small night values.
- GBM on log y alone: best MAPE (97%) but RMSE 264. Averaging the two keeps most of both.
- Bias correction (add the mean past error): RMSE barely moved, MAPE got much worse.
- Left in the errors: the midday peak is under-forecast, and errors persist through the day.
- Next: model the solar and the wind parts separately, and use quantile regression for the interval.